Exploring the Feasibility of Full-Body Muscle Activation Sensing with Insole Pressure Sensors

PENNSTATE 1855

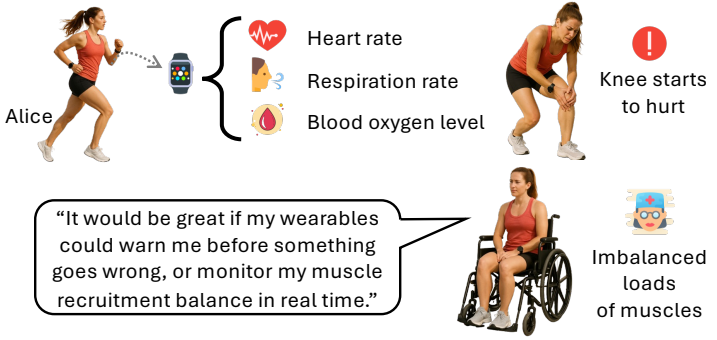
Hao Zhou,
Mahanth Gowda

THE UNIVERSITY OF TEXAS AT DALLAS UTD EST. 1969

Motivation

Knowing Muscle Usage in Daily Life Could Provide More Health Insights

Current sensing methods require skin-mounted electrodes or cameras



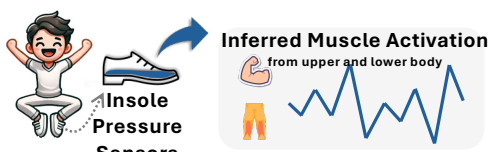
Our Solution

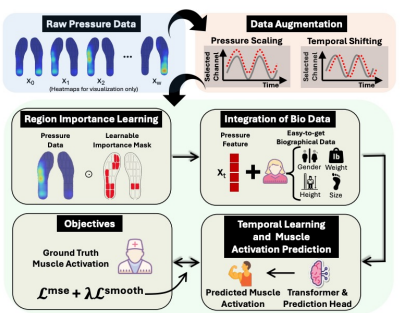
Press2Muscle: Learning Muscle Activation from Foot Pressure Signals

Leveraging biomechanical connections to infer muscle activation from foot pressure

$$\underbrace{\mathbf{a}}_{\text{Muscle Activation}} \rightarrow \boldsymbol{\tau} \rightarrow \ddot{\vec{x}}_{\text{CoM}}, \dot{\vec{x}}_{\text{CoM}} \rightarrow \text{CoP} \rightarrow \underbrace{\mathbf{P}}_{\text{Foot Pressure Pattern}}$$

Subtle motions



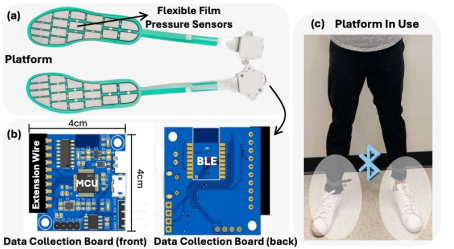


Platform & Evaluation

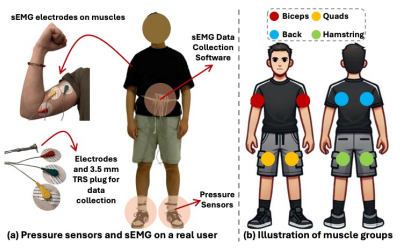
Press2Muscle is Evaluated on A Diverse Cohort

30 Participants, 15 Motions, 8 Muscle Groups, 2 Case Studies

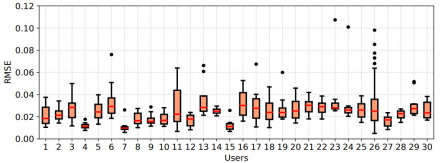
(a) Platform and Setup



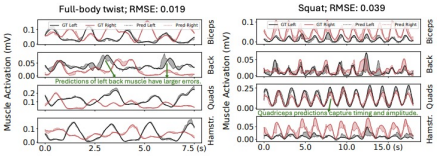
(b) Ground Truth Collection



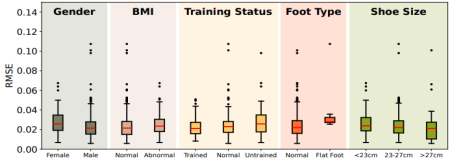
(c) Generalized to Unseen Users



(d) Qualitative Results



(e) Robustness



(f) Muscle Usage Imbalance Free-living

